

Assignment 1 : Linux Boot Process & Directory System

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Deadline : 14th November, 2025 (11:59 PM)

Objectives:

The objective of this assignment is to inspire curiosity in students to explore the boot process and encourage them to dig deeper into how things work underneath the hood — at the lower-level layers before the operating system comes into play. Students will be able to identify and differentiate between different phases of the boot process and will learn about firmware, BIOS, UEFI, GRUB, the kernel, and system processes. Students will also get a better understanding of the difference between user space and kernel space. This understanding will enable them to appreciate the operating systems course more thoroughly and find answers to the fundamental question: What really happens when you press the power button on your PC?

In addition, this assignment also aims to help students build a solid understanding of the Linux directory system — the logical structure that organizes everything within a Linux environment. By exploring the root directory and its subdirectories, students will learn where different components, configurations, binaries, and files reside. This will strengthen their foundation in Linux and make them more confident while navigating and managing file systems.

Program Learning Outcomes (PLOs):

- PLO1: Academic Education
- PLO2: Knowledge for Solving Computing Problems
- PLO3: Problem Analysis
- PLO4: Design/Development of Solutions
- **PLO5: Modern Tool Usage**
- PLO6: Individual and Team Work
- PLO7: Communication
- PLO8: Computing Professionalism and Society
- PLO9: Ethics
- PLO10: Life-long Learning

CLO–PLO Mapping with BT Levels

CLO	Learning Outcome	BT Level	Mapped PLOs
CL01	Practice Linux commands and utilities to gain proficiency with the command-line interface and use it effectively for system management tasks, basic scripting, software installation, and other essential operations in a Linux environment.	A5	PLO5

Section I

Linux Boot Process

Go over the internet and explore everything you can about the **Linux Boot Process**. Create a detailed report covering all the intricacies of every phase involved. Explain the complete process step by step, and try to include as many visuals, animations, pictures, diagrams, or workflows as possible to make your report engaging and to enhance my comprehension of the content.

I'm not specifying any fixed format for your report, you can choose the format as per your own preferences. I expect you to explain everything from scratch — discuss Firmware, Hardware, CPU, BIOS/UEFI, Boot Loader, GRUB, Kernel, Systemd process, and anything else that you consider part of the boot process. Your goal should be to create a complete, well-rounded picture of how Linux boots up.

Section II

Linux Directory System

Similarly, create a comprehensive report explaining the **directory system in a Linux environment**. So far, we have discussed the **root directory (/)**, which serves as the starting point for the entire directory system, and we have explored some of its child directories. We know that the Linux directory system is maintained as a **tree structure**, so in this task, I want you to educate me on this tree — describe all the important directories and files within it.

Use visuals to represent this tree-like structure clearly, and even use smaller sub-trees to explain significant subdirectories or files within them. I am not specifying which directories or files to cover — you are encouraged to explore the Linux filesystem on your own. Normally, I would discourage using chatbots when learning Linux commands, but in this case, I encourage you to use the internet freely: watch videos, read articles, and even consult chatbots to expand your understanding — and, in turn, help me learn from your findings.

Again, I am not prescribing any particular format for your report; you may choose one that suits your taste and creativity. Your peers will naturally raise the bar with their efforts, so make sure to compete by exploring the topic deeply and presenting your findings in a clear and thoughtful manner.

Submission Guidelines

Submit your work as a single PDF report, ensuring that Section I and Section II are clearly separated, either with distinct headings or a title page between the two sections.

Avoid simply copying and pasting AI-generated content; you should understand and own what you write. The authenticity and tone of your writing will matter far more than the length of your report.

I encourage you to **attach links to any videos, articles, or papers** that you found particularly helpful while preparing your report. This will help me understand your learning sources and might also help others explore useful materials later.

I will conduct viva sessions to evaluate your understanding of your report, so make sure everything you include is something you can explain confidently. Only write what you truly understand and can take ownership of.

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Late submissions will not be accepted